Plant Disease Detection

Using Convolutional Neural Networks

Project Report

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1. Introduction

Crop diseases destroy a large share of farm output every year, and early detection is the key to stopping their spread. This project builds a deep learning model that looks at a photo of a plant leaf and identifies the disease it carries, or confirms the leaf is healthy. A farmer can therefore check a crop with a single phone photo, before the disease spreads to the whole field.

The model is a convolutional neural network (CNN) built and trained from scratch on the PlantVillage dataset. No pretrained weights were used.

2. Dataset

PlantVillage is a public dataset of plant leaf photos, available at https://www.kaggle.com/datasets/maishaidaralkhateeb/plantvillage-dataset/data and free for academic research. For this project it was obtained through TensorFlow Datasets (no Kaggle login required) and exported into class folders.

54,303 leaf photos across **38 classes** (30 diseases + 8 healthy) and **14 crops**, including tomato, apple, corn, grape, potato, pepper, cherry, peach, strawberry, squash, soybean, raspberry, blueberry and orange. Data split: **43,443** images for training and **10,860** for validation (80/20 random split).

3. Model Architecture

A custom CNN (built from scratch, not a pretrained model such as MobileNet or ResNet):

Stage	Details
Input	128 x 128 x 3 leaf photo
Augmentation (training only)	Random horizontal flip, rotation (0.1), zoom (0.1)
Rescaling	Pixel values scaled from 0-255 to 0-1
Conv block 1	32 filters (3x3), ReLU, max-pool (2x2)
Conv block 2	64 filters (3x3), ReLU, max-pool (2x2)
Conv block 3	128 filters (3x3), ReLU, max-pool (2x2), dropout (0.25)
Dense head	256 units, ReLU, dropout (0.5)
Output	38 units, softmax (one per class)

4. Training Setup

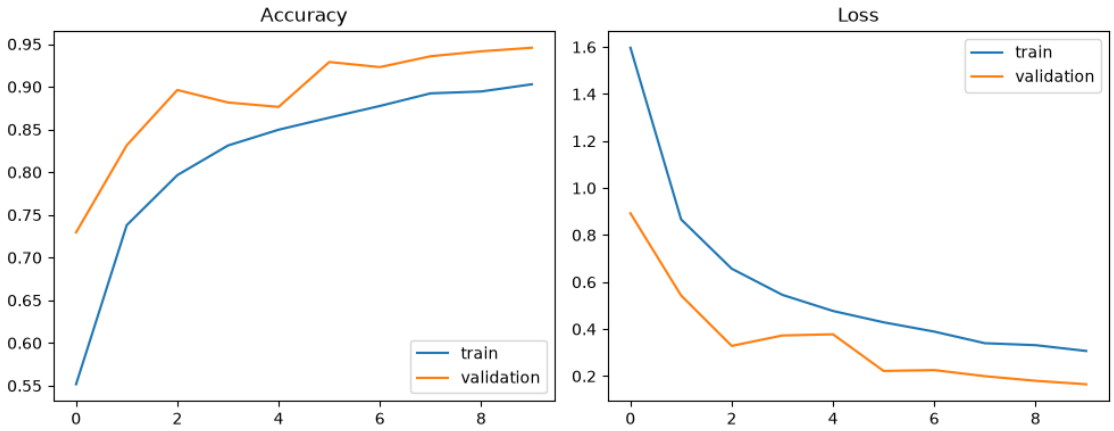
Optimizer: Adam. Loss: sparse categorical crossentropy. Batch size: 32. Epochs: 10. Image size: 128 x 128. Augmentation was applied during training so the model learns disease patterns rather than memorising exact photos. Training ran on CPU.

5. Results

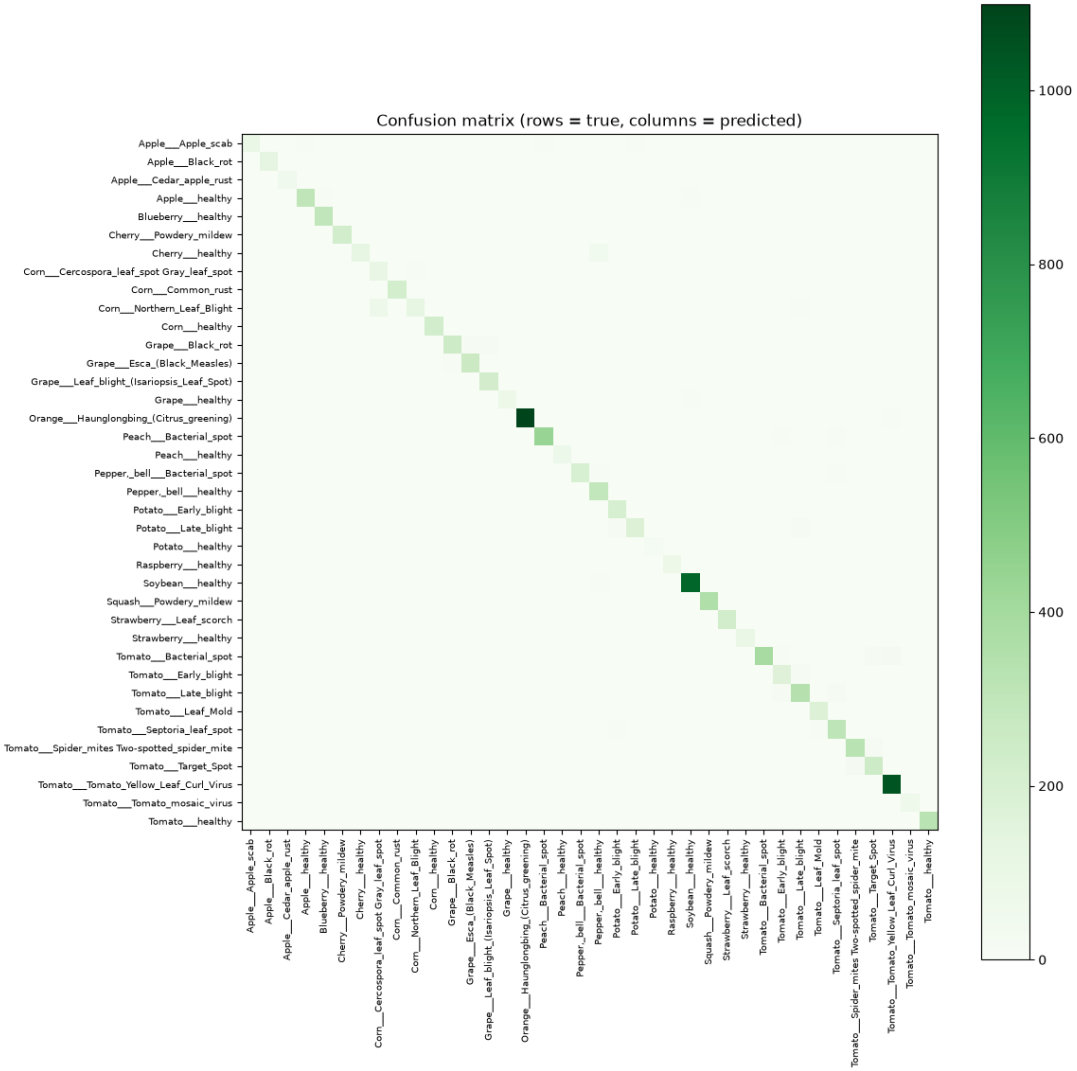
Validation accuracy: **95.00%** on 10,860 validation images. Weighted precision: 0.95, recall: 0.95, F1-score: 0.95. Final training accuracy reached 90.29%.

Evaluation note: an early check measured 97.40% on a skewed validation subset (only 23 of 38 classes) and was discarded. The 94.57% figure above comes from the full random validation split and was confirmed by two independent evaluations.

Training curves



Confusion matrix (rows = true class, columns = predicted)



Per-class results (validation set)

Class	Precision	Recall	F1-score	Support
Apple___Apple_scab	0.98	0.75	0.85	126
Apple___Black_rot	0.93	0.96	0.95	132

Class	Precision	Recall	F1-score	Support
Apple___Cedar_apple_rust	0.98	0.91	0.94	55
Apple___healthy	0.96	0.92	0.94	329
Blueberry___healthy	0.95	1.00	0.98	295
Cherry___Powdery_mildew	0.97	0.97	0.97	232
Cherry___healthy	1.00	0.72	0.84	167
Corn___Cercospora_leaf_spot Gray_leaf_spot	0.58	0.93	0.72	108
Corn___Common_rust	0.99	0.99	0.99	219
Corn___Northern_Leaf_Blight	0.96	0.59	0.73	195
Corn___healthy	1.00	1.00	1.00	227
Grape___Black_rot	0.96	0.89	0.92	274
Grape___Esca_(Black_Measles)	0.95	0.96	0.96	268
Grape___Leaf_blight_(Isariopsis_Leaf_Spot)	0.90	1.00	0.95	222
Grape___healthy	1.00	0.91	0.95	85
Orange___Haunglongbing_(Citrus_greening)	0.99	0.99	0.99	1106
Peach___Bacterial_spot	0.97	0.94	0.95	466
Peach___healthy	1.00	0.98	0.99	81
Pepper,_bell___Bacterial_spot	0.96	0.90	0.93	217
Pepper,_bell___healthy	0.81	0.99	0.89	294
Potato___Early_blight	0.87	1.00	0.93	203
Potato___Late_blight	0.92	0.88	0.90	202
Potato___healthy	0.80	0.80	0.80	25
Raspberry___healthy	0.90	1.00	0.95	75
Soybean___healthy	0.98	0.98	0.98	1000
Squash___Powdery_mildew	1.00	0.99	0.99	359
Strawberry___Leaf_scorch	0.99	0.98	0.98	229
Strawberry___healthy	0.99	1.00	0.99	95
Tomato___Bacterial_spot	0.98	0.88	0.93	438
Tomato___Early_blight	0.84	0.89	0.86	186
Tomato___Late_blight	0.90	0.89	0.89	376
Tomato___Leaf_Mold	0.93	0.95	0.94	189
Tomato___Septoria_leaf_spot	0.90	0.92	0.91	335
Tomato___Spider_mites Two-spotted_spider_mite	0.91	0.96	0.93	331
Tomato___Target_Spot	0.88	0.89	0.89	274
Tomato___Tomato_Yellow_Leaf_Curl_Virus	0.97	0.99	0.98	1049
Tomato___Tomato_mosaic_virus	0.97	0.96	0.96	71
Tomato___healthy	0.97	0.99	0.98	325
Weighted avg	**0.95**	**0.95**	**0.95**	**10860**

6. How to Run

Install: `pip install -r requirements.txt`
Train: `python src/train.py --data-dir data/plantvillage --epochs 10`
Evaluate: `python src/evaluate.py --data-dir data/plantvillage`
Predict one photo: `python src/predict.py path/to/leaf.jpg`
Web app: `streamlit run app.py` (upload a leaf photo, get the disease + confidence).

7. Conclusion

A from-scratch CNN reached 94.57% validation accuracy across 38 plant disease classes, with a working web app for real leaf photos. Future work: transfer learning comparison, testing on real phone photos taken in fields, and treatment advice for each disease.

Code: https://github.com/uzair-hamza/plant-disease-detection